Visual Recognition using Deep Learning
2026 Spring, Homework 3

Release Date: 2026/04/21 15:40

Homework 3

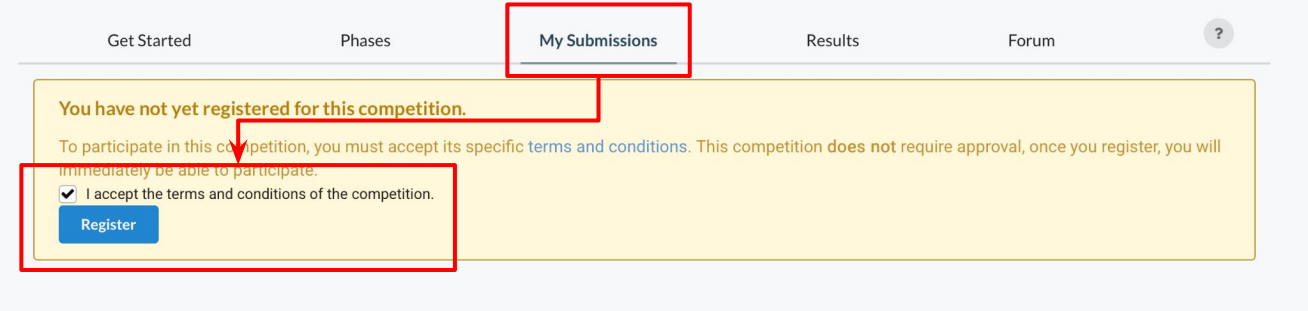
- **Deadline:** 23:59, 05/12, 2026
- **Participate the competition** (85%): Instance Segmentation
  - Participant the competition on the CodaBench and get the highest score as possible. (80%)
  - Code reliability & quality: GitHub (5%)
- **Report and code** (15%): Document your method and findings.
  - Report
    - In PDF format and written in English. (5pt penalty)
    - Introduction to your method (e.g., data pre-processing, model architecture, hyper-parameters)
    - Innovative ideas or additional experiments to further improve the model.
  - Code
    - Zip your code (.py) alone with report - Submit to E3.
    - You should also put your code on your GitHub repository and provide the link in the report.

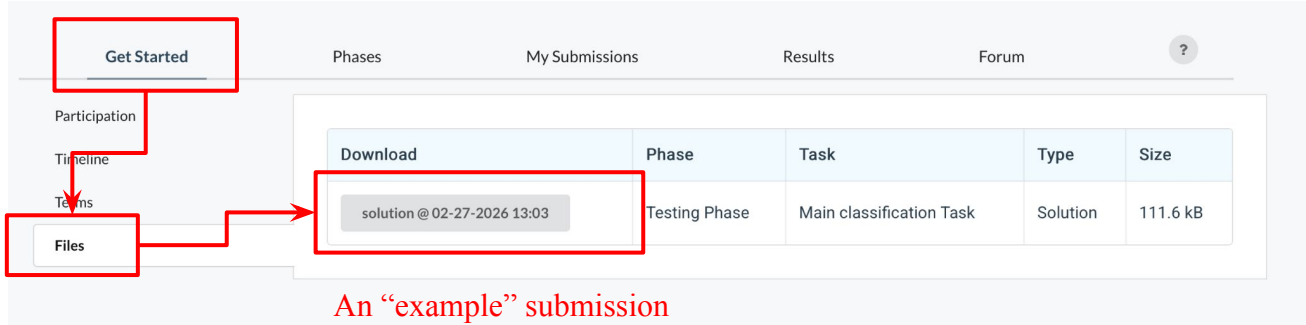
Links

- Link to the dataset
- Link to the competition
- Link to sample code

How to participate the competition and do submission

1. Register an account on CodaBench. (**Do not** use two accounts to participate in the competition.)
2. After you click the competition link, go to My Submissions, and join the competition.

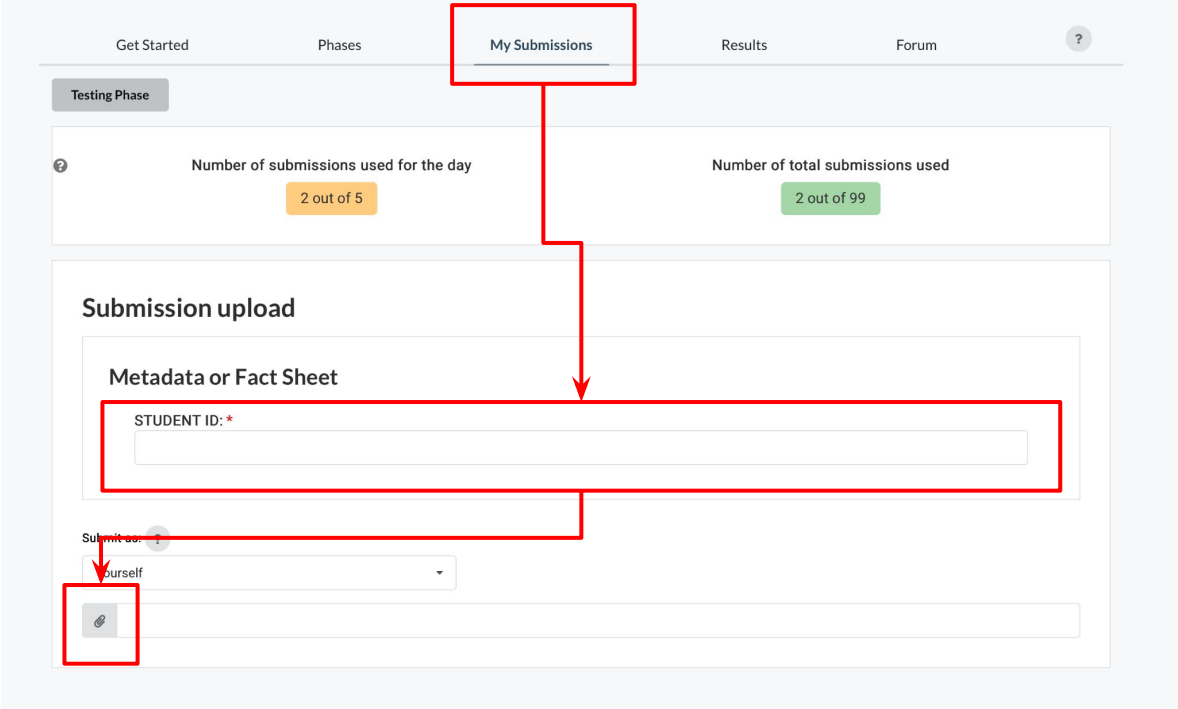




An "example" submission

How to participate the competition and do submission

3. Enter your student ID number, then upload the submission file.

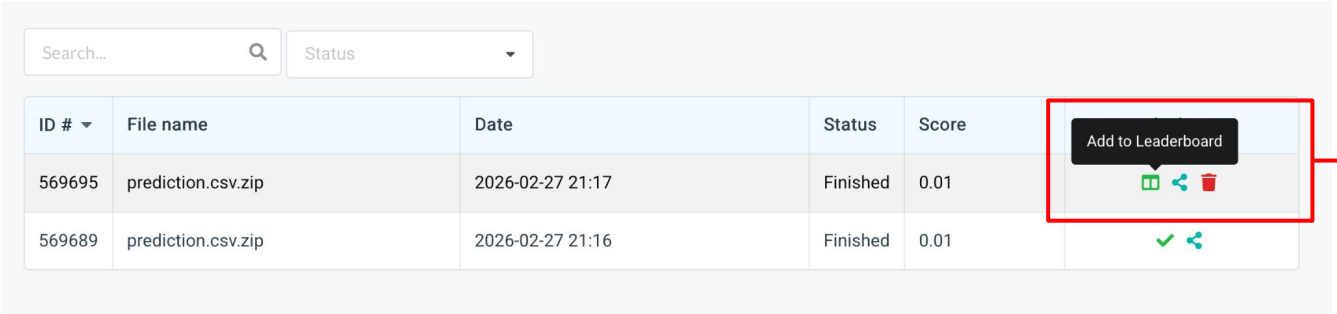


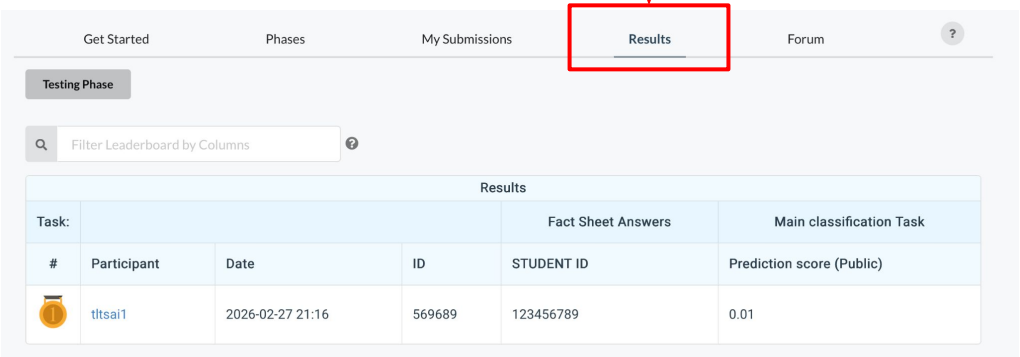
No limitation for the name of the "zip" file; however, inside the zip file, your **result file must be named "test-reults.json"**

How to participate the competition and do submission

4. Don't forget to "Add to Leaderboard".

5. Don't forget to check your results on the leaderboard.





Coding Environment

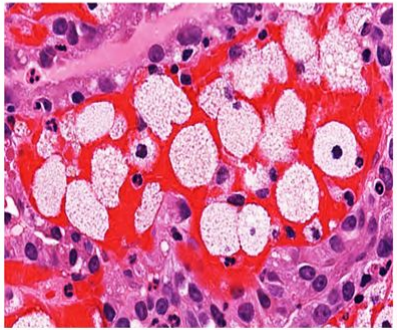
- Recommnedation: Python 3.9 or higher
- Tips
  - We recommend you to use **virtual environments** when implementing your homework assignments.
  - Here are some popular virtual environment management tools
    - Poetry
    - Conda
    - Virtualenv

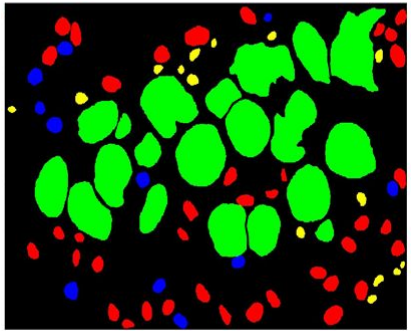
Numpy & PyTorch

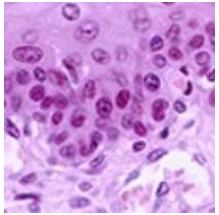
- Numpy Tutorial: Link
- PyTorch Tutorial: Link
  - Free to use any modules and functions

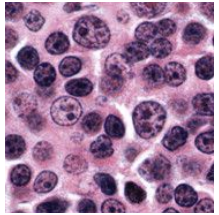
Task and Dataset

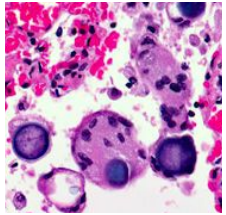
- Task type: Instance Segmentation
- Dataset
  - Colored Medical Images
    - Training / Validation: 209 images
    - Test: 101 images
- Target
  - Segmentation masks of 4 types of cells (class1, class2, class3, class4)
- Evaluation
  - AP 50

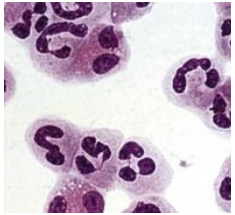












Task Requirement and Limitations

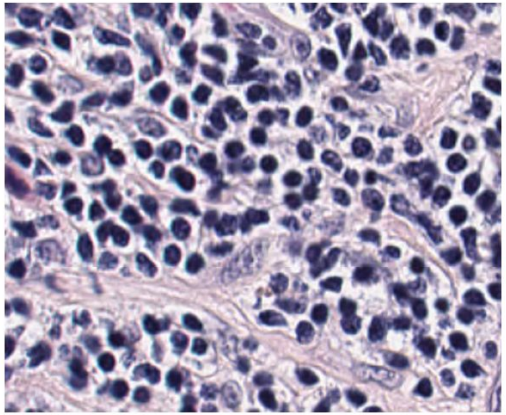
- **Requirement**: Train instance segmentation models to segment target objects.
  - We provide you the raw image and masks in .tif format.
    - You need to process the data by yourself.
    - You need to convert predicted masks into correct submission format (example).
- **Limitations**
  - No external data (i.e., data from other sources) is allowed.
  - Only pure vision-based model is allowed (No vision-language based model; No prompt-based model)
  - You can base on Mask R-CNN to modify components/modules to improve the model performance.
    - You must "elaborate" (i.e., key design/contribution of that work) and "cite" the paper in the report.
    - Your model size (trainable parameters) should less than **200M**.
- **Note**: Pretrained weights (ImageNet) is allowed.

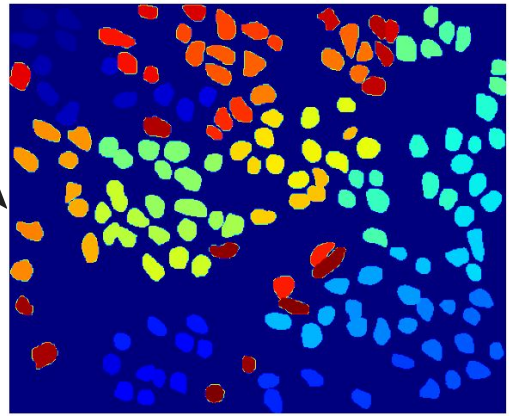
Dataset Inspection

train/
- [image_name]
  - image.tif
  - class1.tif
  - class3.tif
  - …

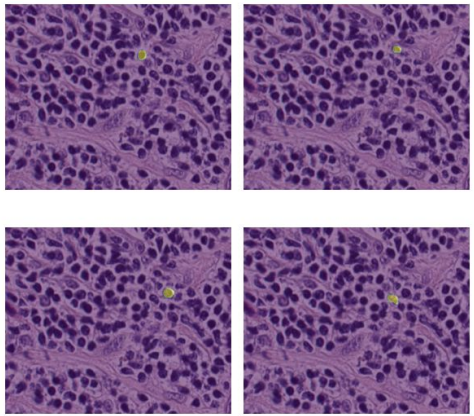
test/
- [image_name].tif

test_image_name_to_ids.json





Each unique pixel value represents an instance.
(e.g., mask == 1, mask == 2, mask == 3, mask == 4)




```
[{
    "file_name": "c8cb7626-7423-4c1e-a81c-5ff25ea180b3.tif",
    "id": 1,
    "height": 446,
    "width": 512
},
{
    "file_name": "8bf17017-577c-4bc7-b599-df8289a69279.tif",
    "id": 2,
    "height": 151,
    "width": 147
},
...
]
```


For you to generate submission file, you will need to map the filename to image_id

Output Format Example for Instance Segmentation

- Official instruction
- What is RLE format
- Sample code for you to encode mask to RLE format
- To prevent blind submission - Run evaluation by yourself before submitting (see COCOEval)

For detection with bounding boxes, please use the following format:

```
[{
  "image_id"          : int,
  "category_id"       : int,
  "bbox"              : [x,y,width,height],
  "score"             : float,
}]
```

```
[{
  "image_id"          : int,
  "category_id"       : int,
  "segmentation"      : RLE,
  "score"             : float,
}]
```


```
[
    {
        'image_id': 1,
        'bbox': [95.22177124023438, 381.214111328125, 24.7103271484375, 25.109375],
        'score': 0.56789,
        'category_id': 1,
        'segmentation': {
            'size': [446, 512],
            'counts': 'PhY1f0W=1000O100O2N1O2N1N3N2M5Jgb_5'
        }
    },
    {
        'image_id': 1,
        'bbox': [304.9966735839844, 241.36700439453125, 45.23297119140625, 41.11309814453125],
        'score': 0.45678,
        'category_id': 1,
        'segmentation': {
            'size': [446, 512],
            'counts': 'cRU4d0Y=3N1O01O01O10O010O10O01O01O0001O00O100O101N1O2M3N2M5JjT^2'
        }
    }
    ...
]
```

Grading Policy - Report (15%)

- Format: PDF, written in English. (-5pts if not followed)
- Sections that you should include
  - **Introduction** to the task and core idea of your method
  - **Method:** Describe your data preprocessing procedure, model architecture, hyperparameter settings, and any modifications or adjustments you made.
    - You need to describe each key component in your model. (e.g., Backbone: ResNet, Neck: FPN, Heads: Mask/Transformer decode, etc.)
  - **Results**: your findings / model performance / visualization (e.g., training curve, confusion matrix, etc.)
  - **References**: Your method references (paper / Github sources, must include if you use any). You may refer to the ECCV 2026 template for the citation and reference formatting guidelines.

12pts

We encourage you to stand on the shoulders of giants - only clone and run it is not enough.

3pts

- **Additional experiments** to explore better performance
  - Simply tuning the hyper-parameters doesn't count (e.g., batch-size, LR, different optimizers).
  - Hint: Try to add/remove some layers, use different loss functions, etc.
- You should 1) include your hypothesis (why you do this), 2) How this may (or may not) work, and 3) The experiment results and their implications.

Grading Policy - Code Reliability (5%)

1. Please follow the PEP8 instructions and lint your code.
2. Push your code to the GitHub
   - It should contain a README.md file that includes the following sections:
     - Introduction
     - Environment Setup
     - Usage
     - Performance Snapshot

Python Coding Style Guide Reference

1. [PEP8]
2. [Google Python Style]

NYCU Computer Vision 2026 HW1

- **Student ID:** 123456789
- **Name:** Your Name

Introduction

Briefly introduce your work here.

Environment Setup

How to install dependencies.

```
pip install -r requirements.txt
```

Usage

Training

How to train your model.

```
python train.py
```

Inference

How to run inference.

```
python inference.py
```

Performance Snapshot

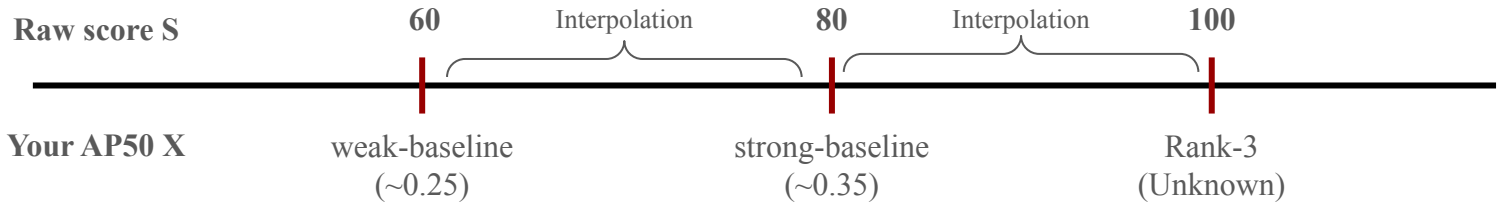
Insert a screenshot of the leaderboard here.

Grading Policy - Competition (80%)

We will use **private (hidden) leaderboard** to evaluate the performance (the distribution is similar for data in public and private set.) The public leaderboard is for you as reference.

Your score (competition):

- **Less than weak-baseline:** S = 0
- **Between weak-baseline and strong baseline:** 60 + (X - w.baseline) / (s.baseline - w.baseline) * (80 - 60)
- **Between strong-baseline and Rank3:** 80 + (X - s.baseline) / (rank3 - s.baseline) * (100 - 80)
- **Rank1, 2, 3:** 100



Submission

- Compress your **code** and **report** into a **.zip file** and submit it to E3.
  - Don't forget to push your code to GitHub. And your GitHub link should be written in the report.
- Report should be written in English.
- <STUDENT ID>_HW3.zip
  - codes (.py, folders, etc)
  - <STUDENT ID>_HW3.**pdf** (NO .doc, .docx or others format)
- Don't put the data (e.g. x.jpg / train.csv / test.csv) and model checkpoints into submission file. (-5pts if not followed)

Other rules

- **Late Policy**: A penalty of **20 points** per additional late day. (-20pts per delayed day)
  - For example, If you get 90 points but delay for two days, your will get only 50 points!

- **No Plagiarism**: You should complete the assignment by yourself. Students engaged in plagiarism will be penalized heavily. Super serious penalty.
  - e.g. -100pt for the assignment or failed this course, etc
  - Report to academic integrity office

FAQs

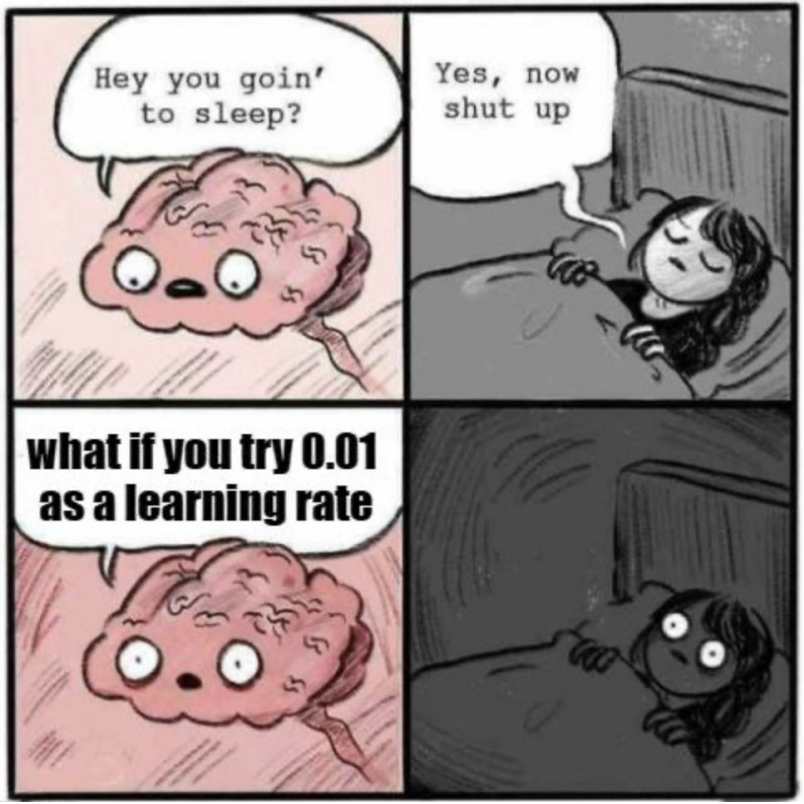
- Can I use any library/package/framework from GitHub or other resources?
  - **Yes**, we encourage you to learn how to leverage existing knowledge on your own task
    - e.g., Github of published works and model zoo from Torchvision
    - Focus on how to step forward from them - That's why part of scores comes from your competition ranks
  - You **should not copy-and-paste from your classmates** (Plagiarism)
- How to handle the GPU Out-of-Memory (OOM) issue?
  - Easy answer - Make your batch size smaller or make your model smaller.
  - Advanced methods: Try to figure it out by yourself. (Many online resources and AI-assistance)

FAQs

- If I don't have my own GPU – Use Google Colab
  - It should be 12 hours, please check this discussion in the stackoverflow
  - And some tricks here may make it longer.

- If you have other questions, ask on **E3 forum** first! We will reply as soon as possible.

It’s your turn! Have Fun!



Team-up for the final project!

Deadline: 05/03 (Sun). After 05/04, we will random assign

Find 3-4 classmates to team up. [Link to the form]

- "Team Member 1" will be the leader (We'll contact leader when needed.)
- Feel free to invite/join using E3 discussion board. (Just use homework discussion board)

Report order may be related to "topic" and in a random order - announce after the topic is selected.

A	B	C	D	E	F	G
GroupID	GroupName	TeamMember1	TeamMember2	TeamMember3	TeamMember4	Selected Topic
Group1						
Group2						
Group3						
Group4						
Group5						
Group6						
Group7						
Group8						
Group9						
Group10						

[Student ID], [Your Name]

Group Name
Can be either Chinese English
No affect to the grading

Topic will be released in the future.